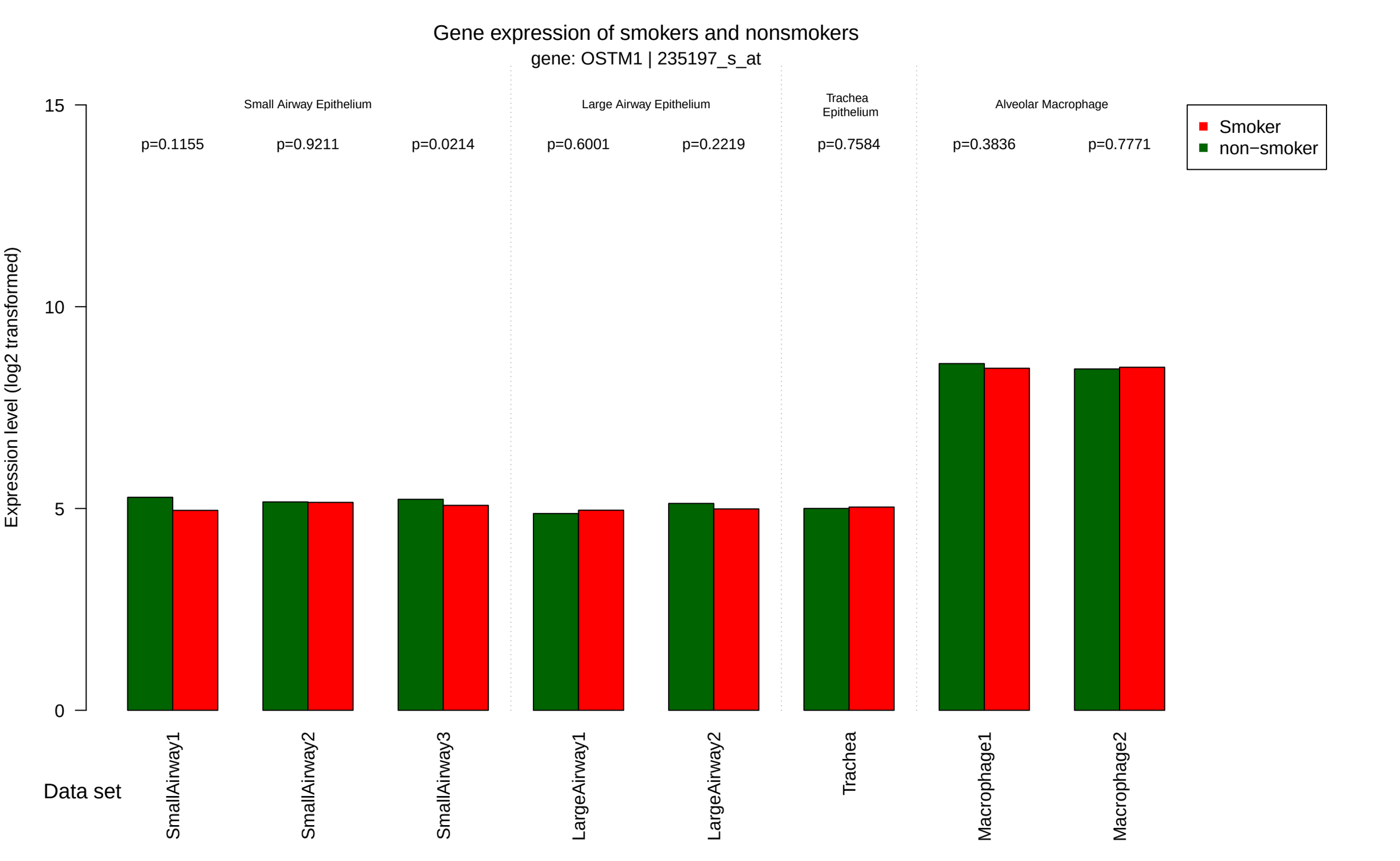
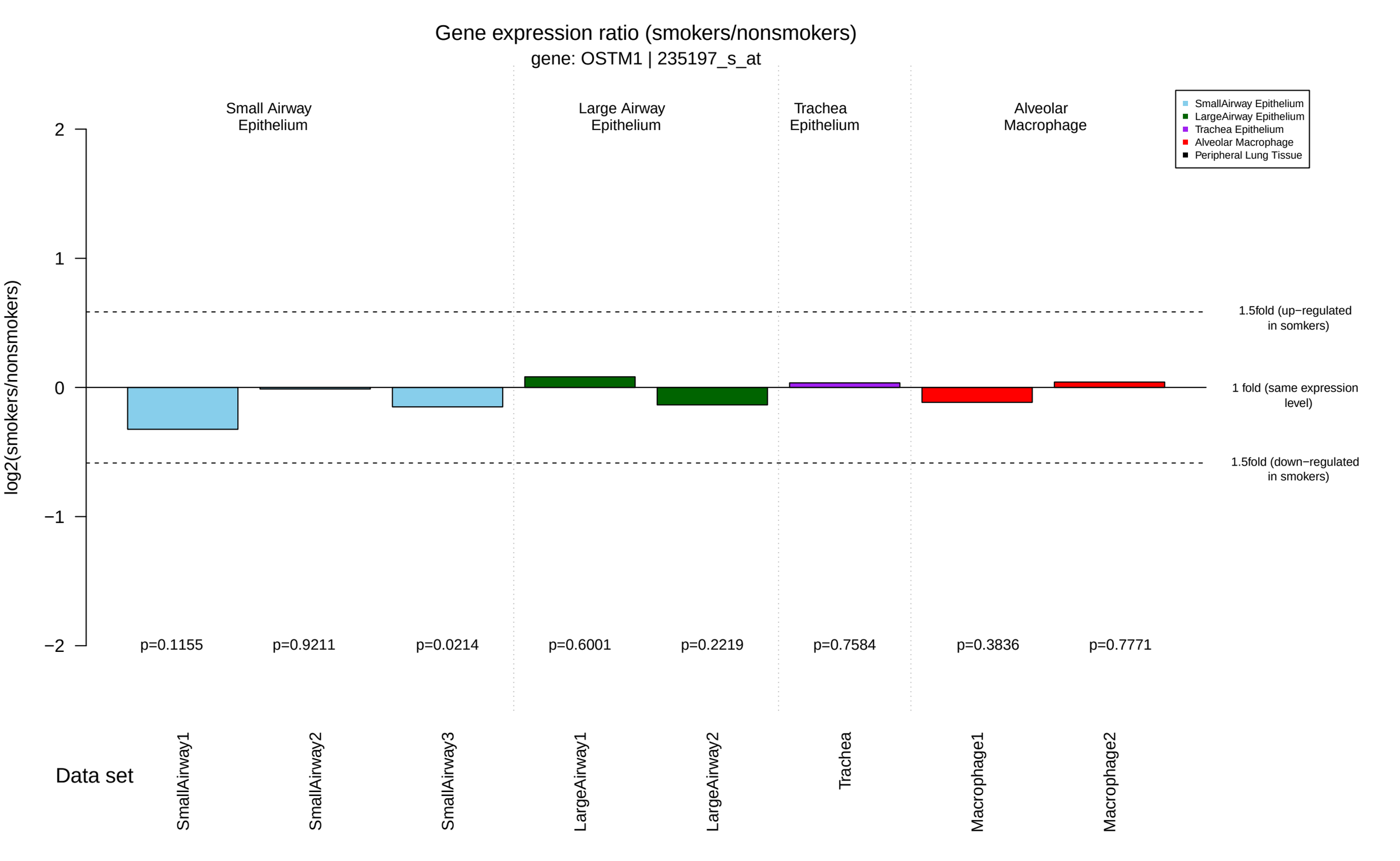
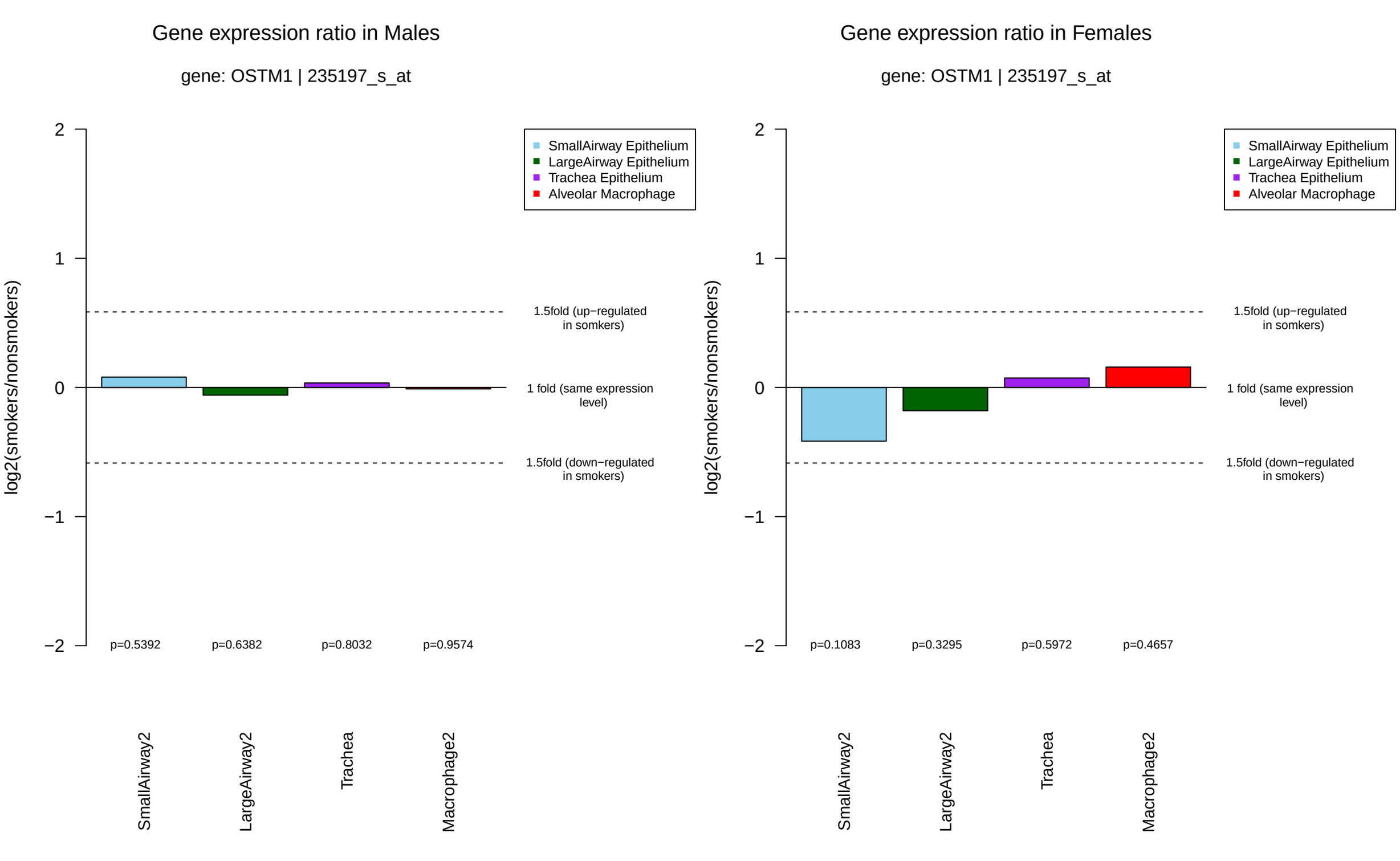
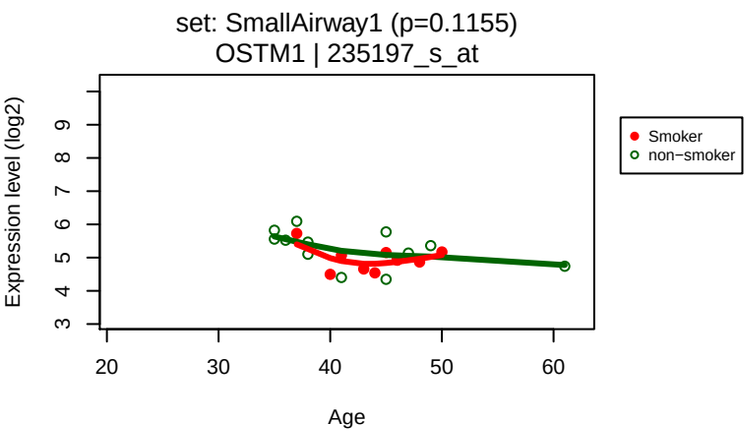
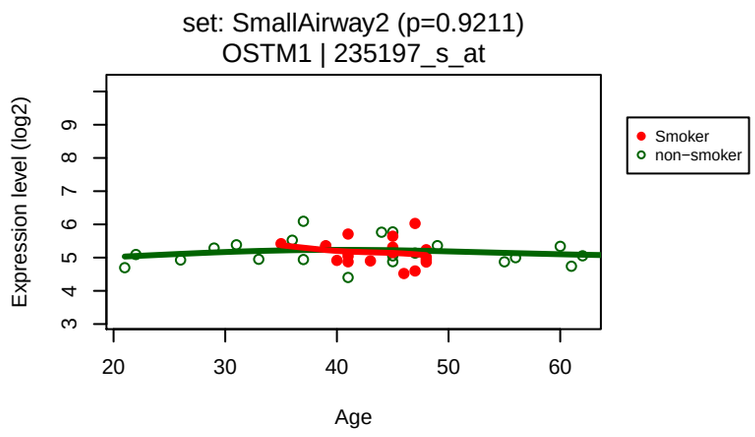
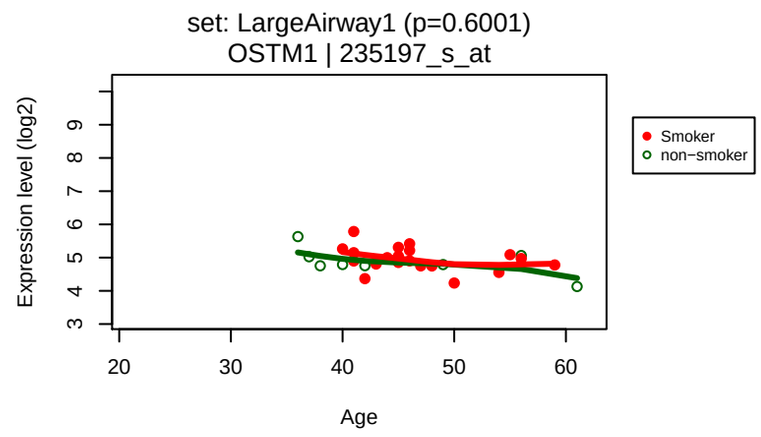

Gene expression of smokers and nonsmokers
gene: OSTM1 | 235197_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1155
p=0.9211
p=0.0214
p=0.6001
p=0.2219
p=0.7584
p=0.3836
p=0.7771
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

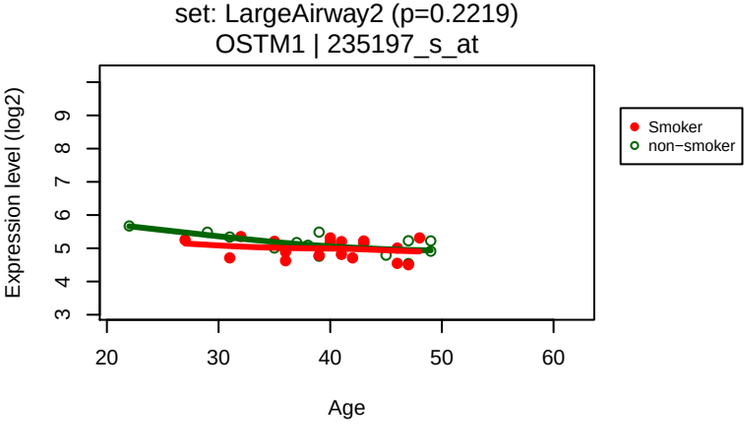

Gene expression ratio (smokers/nonsmokers)
gene: OSTM1 | 235197_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1155
p=0.9211
p=0.0214
p=0.6001
p=0.2219
p=0.7584
p=0.3836
p=0.7771
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

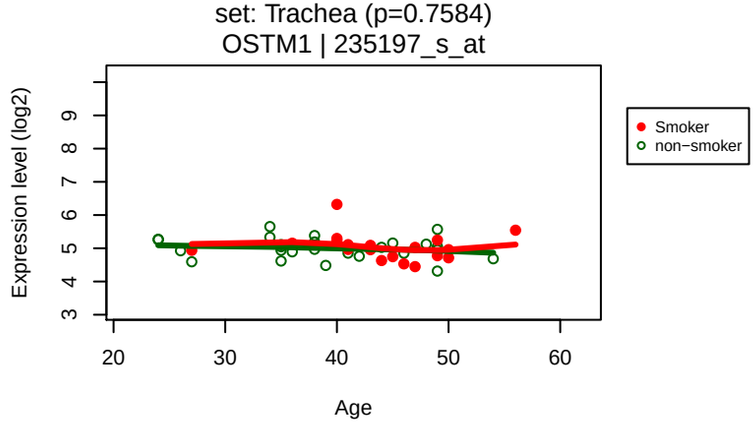

Gene expression ratio in Males
gene: OSTM1 | 235197_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5392
p=0.6382
p=0.8032
p=0.9574
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OSTM1 | 235197_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1083
p=0.3295
p=0.5972
p=0.4657
SmallAirway2
LargeAirway2
Trachea
Macrophage2

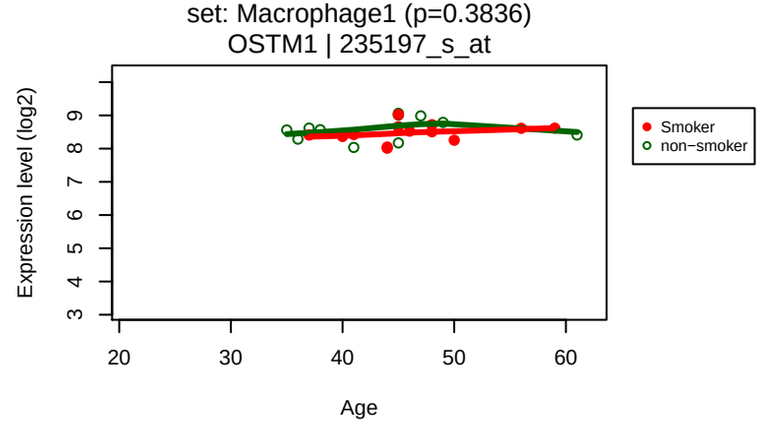

set: SmallAirway1 (p=0.1155)
OSTM1 | 235197_s_at
Expression level (log2)
Age
Smoker
non−smoker

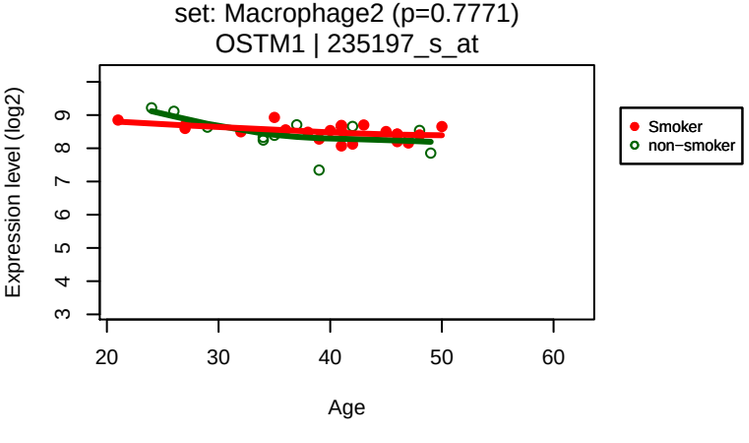

set: SmallAirway2 (p=0.9211)
OSTM1 | 235197_s_at
Expression level (log2)
Age
Smoker
non−smoker

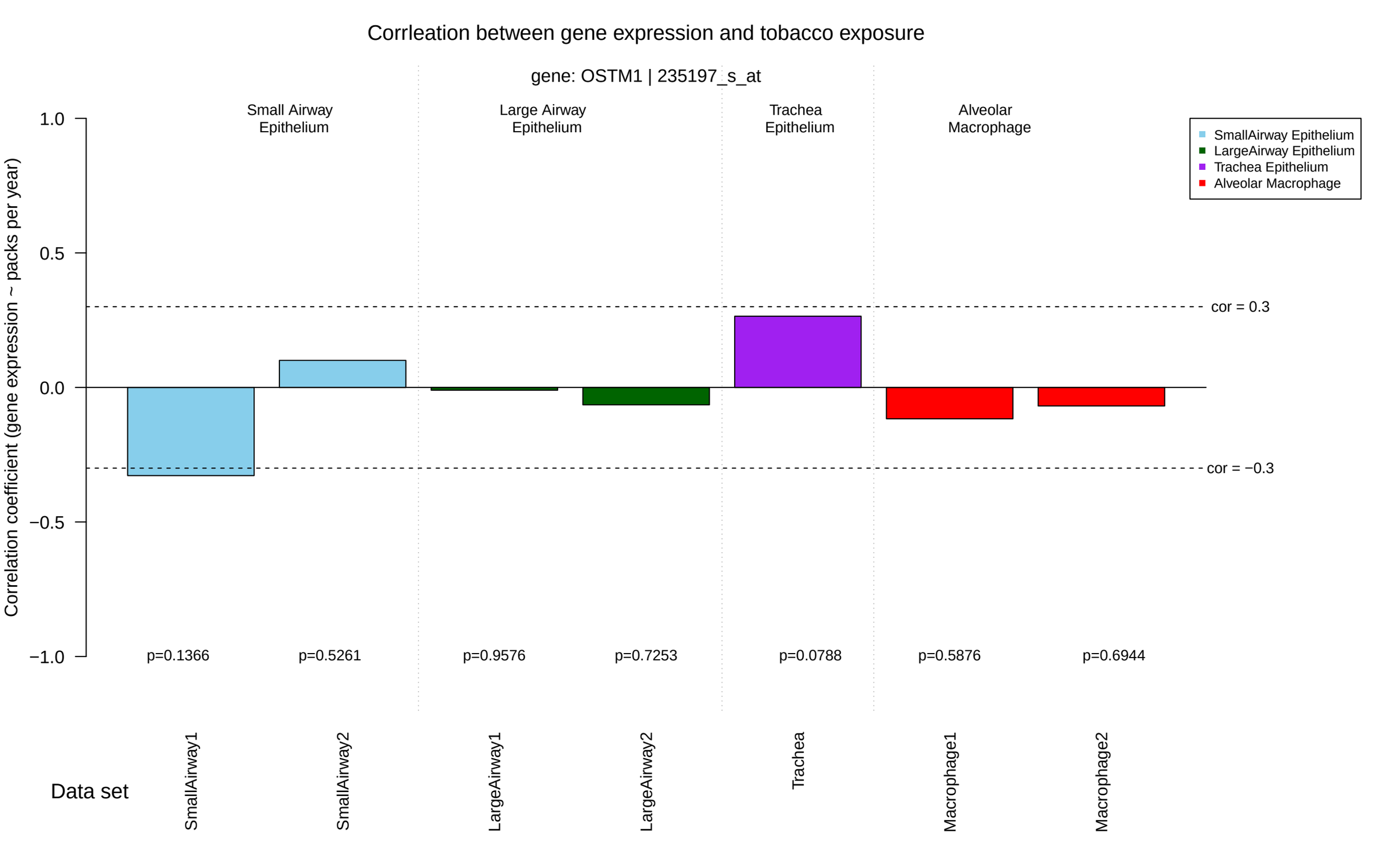

set: LargeAirway1 (p=0.6001)
OSTM1 | 235197_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2219)
OSTM1 | 235197_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7584)
OSTM1 | 235197_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3836)
OSTM1 | 235197_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7771)
OSTM1 | 235197_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OSTM1 | 235197_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1366
p=0.5261
p=0.9576
p=0.7253
p=0.0788
p=0.5876
p=0.6944
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2